FACT: Fact-Augmented Claim Tester for Detecting Hallucination in LLMs

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Introduction

- Large Language Models (LLMs) have become widely adopted for their ability to generate human like text with reasoning capability.
  - However, they are still prone to hallucinations.
- This project presents a fact-checking pipeline designed to detect hallucinated claims in LLM outputs by retrieving transparent evidence to verify these claims.

Background

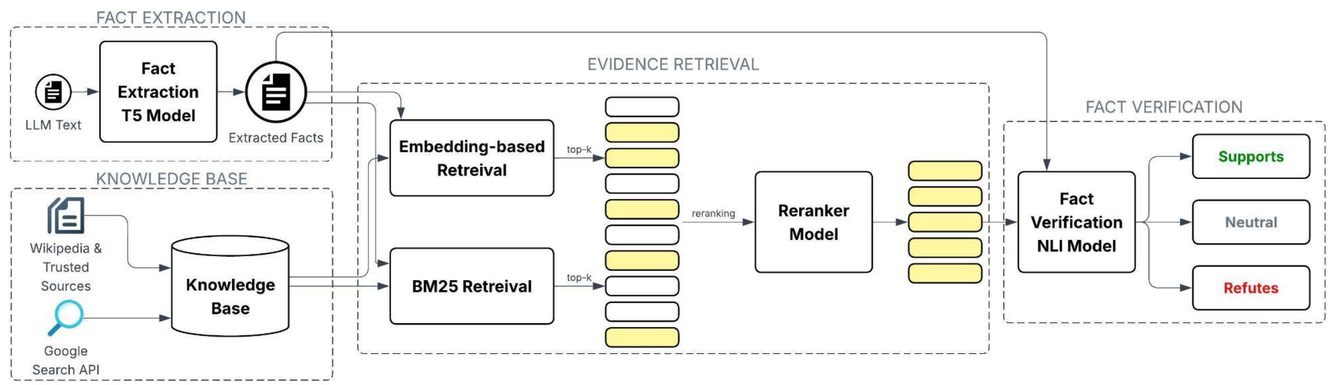
- Research by OpenAI shows that their reasoning models such as OpenAI o3 and o4-mini hallucinate more than their previous models.

Table 4: Hallucination evaluations

Dataset	Metric	o3	o4-mini	o1
SimpleQA	accuracy (higher is better)	0.49	0.20	0.47
	hallucination rate (lower is better)	0.51	0.79	0.44
PersonQA	accuracy (higher is better)	0.59	0.36	0.47
	hallucination rate (lower is better)	0.33	0.48	0.16

- Solutions such as Retrieval-Augmented Generation (RAG) provide information to ground the responses on during generation, but fails to verify the final responses.

System Design



1. Fact Extraction
- Decomposes LLM outputs into atomic, verifiable claims using a fine-tuned Google's Flan-T5 transformer model.

2. Knowledge Base
- Builds the knowledge base by compiling text from trusted truth sources, such as Wikipedia and dynamically updates using Google Custom Search JSON API for niche or recent topics.

3. Evidence Retrieval
- Retrieves supporting evidence from the knowledge base using the two-stage retrieval framework; hybrid retrieval using embeddings and BM25 score retrieves a set of documents that is reranked by a Cross-Encoder model to output a final set of most relevant evidence.

4. Fact Verification
- Verifies each extracted fact with the evidence using a DeBERTa-based natural language inference (NLI) model to classify their label.

Results

Component Evaluation

Stage	Precision	Recall	F1 Score
Claim Extraction	0.673	0.836	0.745
Retrieval Stage 1 (Hybrid)	–	0.74	–
Retrieval Stage 2 (Reranker)	–	0.68	–
Fact Verification (Overalll)	0.759	0.791	0.774
SUPPORTS	0.792	0.816	0.804
REFUTES	0.721	0.760	0.740

Table 1. FACT evaluation metrics for claim extraction, two-stage retrieval, and fact verification. Retrieval stages do not compute precision/F1

End-to-End Evaluation

$$\text{E2E Weighted Score} = \frac{\text{correctly extracted and verified claims}}{\text{total claims}}$$

Class	E2E Weighted Score
SUPPORTS	0.613
REFUTES	0.694
Overall	0.643

Table 2. E2E weighted score per class

Future Work

- more extensive evidence retrieval such as multi-hop and graph retrieval
- improved claim extraction methods to better handle more complex responses.

Examples

"A transformer in machine learning is a neural network architecture designed to handle sequential data—such as text, audio, or time series—without relying on recurrence or convolutions. Instead, it uses a mechanism called self-attention, which allows the model to look at all positions in the input sequence at once and determine which parts are most relevant to each other."

CLAIM: A transformer in machine learning is a neural network architecture designed to handle sequential data without relying on recurrence or convolutions.

Evidence Text: The Transformer architecture follows an encoder-decoder structure but does not rely on recurrence and convolutions in order to generate an output. The task of the encoder, on the left half of the Transformer architecture, is to map an input sequence to a sequence of continuous representations, which is then fed into a decoder. =

Verification Result: SUPPORTS (Score: 0.507)

CLAIM: Self-attention allows the model to look at all positions in the input sequence at once and determine which parts are most relevant to each other.

Evidence Text: As the model processes each word (each position in the input sequence), self attention allows it to look at other positions in the input sequence for clues that can help lead to a better encoding for this word. Self-attention is the method the Transformer uses to bake the "understanding" of other relevant words into the one we're currently processing.

Verification Result: SUPPORTS (Score: 0.821)

"Anne Frank and her sister Margot were found alive in the Bergen-Belsen concentration camp after new research was released by the Anne Frank House. Witnesses at the camp stated that they saw the sisters alive as late as February 7, contradicting the date of death previously believed. The sisters endured terrible living conditions and disease, but ultimately survived the Holocaust."

CLAIM: Anne Frank and her sister Margot were found alive in the Bergen-Belsen concentration camp.

Evidence Text: In October or November 1944, Anne and her sister, Margot, were transferred to Bergen-Belsen concentration camp from Auschwitz, where they died - probably of typhus - a few months later. They were originally estimated by the Red Cross to have died in March, with Dutch authorities setting 31 March as their official date of death, but research by the Anne Frank House in 2015 suggests they more likely died in February.

Verification Result: REFUTES (Score: 0.901)

CLAIM: Witnesses at the camp stated that they saw the sisters alive as late as February 7.

Evidence Text: The disease causes high fever, chills and skin eruptions. "Because of the lice infesting the bedstraw and her clothes, Anne was exposed to the main carrier of epidemic typhus for an extended period," museum researchers wrote. They concluded that it's unlikely the sisters survived until March, because witnesses at the camp said the sisters both had symptoms before February 7.

Verification Result: REFUTES (Score: 0.834)

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